

Project Proposal

Context-Aware Next Best Action Engine Using Simulated App Events

1. One-line Summary

Build a prototype Next Best Action engine using simulated app event data. The system will analyze user behavior, identify user personas, rank the most relevant next actions, explain the recommendations, and show the results through a lightweight dashboard.

The project will not build a full mock super-app. Instead, it will define a lightweight event/action environment that represents common consumer-app journeys such as voucher usage, payments, food/merchant browsing, loyalty rewards, finance education content, and re-engagement.

2. Context

Many consumer apps and super-apps contain multiple services, campaigns, merchants, content, and user journeys. Users may not always discover the most relevant feature, offer, or content at the right time.

This project focuses on the decision engine behind personalization rather than the app itself. The goal is to explore how user behavior signals can be transformed into personas and next-best-action recommendations.

The project uses synthetic data because real app data is not available. It does not claim to solve a validated production problem or improve real conversion. The main learning goal is to understand recommendation, decisioning, feature engineering, evaluation, explanation, and system design in a way that can transfer to super-apps, fintech apps, e-commerce, loyalty systems, and consumer platforms.

3. Problem Hypothesis

In a multi-service consumer-app environment, user behavior signals such as voucher usage, payment activity, merchant views, content reads, session frequency, and inactivity can be used to infer user intent/persona and recommend a more relevant next action.

Example scenarios:

- User often uses vouchers and payments → recommend cashback campaign.
- User reads finance education content but has not used the feature → recommend beginner education content.

- User often opens food, delivery, or restaurant pages around lunch time → recommend a lunch-time food/merchant offer.
- User has low recent activity → recommend re-engagement offer.
- User frequently uses one service → recommend shortcut to that frequent service.
- User collects loyalty points but rarely redeems them → recommend loyalty reward usage.

4. Research Inspiration / Benchmark Patterns

Key design patterns extracted:

1. Two-stage decisioning: candidate action generation → action ranking.
2. User persona / intent modeling: behavior history becomes an interpretable user profile.
3. Context-aware recommendation: time, recency, inactivity, and feature usage affect recommendations.
4. Exploration vs exploitation: recommend relevant actions while still discovering new interests.
5. Eligibility and safety rules: not every high-scoring action should be shown.
6. Explainability and feedback loop: the system should explain recommendations and learn from feedback.

5. Simulated Event & Action Environment

Instead of building a mock super-app UI, the project will define a lightweight simulated environment consisting of:

1. User personas.
2. User event types.
3. App feature categories.
4. Candidate recommendation actions.
5. Feedback signals.
6. Safety and eligibility rules.

This provides enough context for recommendation without spending time building a full fake application.

Example App Feature Categories

- Payments
- Merchant / food discovery
- Voucher and cashback campaigns
- Loyalty rewards
- Finance education content
- Re-engagement and onboarding

Example User Events

- view_voucher
- redeem_voucher
- make_payment
- view_merchant
- open_food_page
- read_finance_education
- click_campaign
- dismiss_recommendation
- use_loyalty_feature
- inactive_for_days

Example Recommendation Actions

- recommend_cashback_campaign
- recommend_lunch_time_food_offer
- show_beginner_finance_education_content
- show_payment_shortcut
- show_reengagement_offer

- show_loyalty_reward
- show_onboarding_tip

6. Proposed Solution

The system will simulate user events, transform them into user-level features, detect personas, generate candidate actions, rank the next best actions, and explain why each action was recommended.

Workflow:

User events

- feature engineering
- persona / intent detection
- candidate action generation
- eligibility and safety filtering
- action ranking
- explanation generation
- feedback logging
- dashboard and evaluation

7. Input Data

Because real app data is not available, the project will create a synthetic event dataset based on designed user personas and behavior patterns.

Example user features:

- voucher_redeemed_30d
- payment_count_30d
- food_page_views_lunch_7d
- finance_content_views_30d
- last_active_days
- session_count_7d
- campaign_click_rate
- recommendation_dismiss_rate
- loyalty_usage_rate

The synthetic data should be generated from behavior patterns, not random noise.

Example personas:

Persona

Expected Behavior

Deal-seeking
active
user

Frequently views/redeems vouchers,
uses payments, clicks campaigns.

Lunch-time
food
explorer

Opens food or merchant pages
around lunch time.

Finance-curious
beginner

Reads finance education content but
has not used finance-related
features.

Dormant user	Previously active but low recent activity.
Payment-heavy user	Uses payment often but rarely uses loyalty or campaign features.
Loyalty-aware user	Collects points but rarely redeems rewards.

8. Output

For each user, the system will return:

- user persona
- top recommended actions
- recommendation score
- explanation

- applied rules or constraints
- feedback log

Example output:

```
1. {  
2.   "user_id": "U123",  
3.   "persona": "deal-seeking active user",  
4.   "top_actions": [  
5.     {  
6.       "action": "recommend_cashback_campaign",  
7.       "score": 0.86,  
8.       "reasons": [  
9.         "User redeemed 5 vouchers in the last 30 days",  
10.        "User made 8 payments in the last 30 days",  
11.        "User has high campaign engagement"  
12.      ]  
13.     },  
14.     {  
15.       "action": "recommend_lunch_time_food_offer",
```

```
16.    "score": 0.72,  
17.    "reasons": [  
18.        "User often opens food or merchant pages around lunch time"  
19.    ]  
20. }  
21. ]  
22. }
```

9. Lightweight Dashboard

The project will include a simple dashboard to demonstrate the system.

The dashboard is not a full mock super-app. It is an inspection and demo tool for the NBA engine.

Dashboard Views

1. User Profile View

- User ID
- Persona

- Key behavior features
- Recent events

2. Recommendation View

- Top recommended actions
- Scores
- Reasons
- Applied eligibility/fatigue/safety rules

3. Feedback Simulation View

- Click
- Dismiss
- Ignore
- Not interested

4. Evaluation View

- Scenario accuracy
- Precision@K
- NDCG@K
- Persona distribution
- Action distribution
- Explanation faithfulness checks

The dashboard helps mentor/reviewer understand the logic without requiring a real app UI.

10. AI / DS / ML / SE / Ops Mapping

Data Science

- Design realistic user personas and behavior scenarios.
- Analyze user behavior from event logs.
- Build features based on recency, frequency, time context, and engagement.
- Segment users into interpretable personas.
- Evaluate recommendation quality using scenario tests and ranking metrics.

Machine Learning

- Start with a rule-based baseline.
- Add clustering for persona discovery.
- Optionally add a simple classification or ranking model to predict the most relevant action.
- Compare rule-based and ML-based approaches.

AI / LLM

- Optional explanation generation.

- The LLM will not decide the recommendation directly.
- It will only explain recommendations based on computed features and scores.
- Guardrails will be added to avoid unsafe financial advice.

Software Engineering

- Design event schema and action catalog.
- Build synthetic data generator.
- Build feature engineering pipeline.
- Build recommendation service/API.
- Build lightweight dashboard.
- Store recommendation and feedback logs.

Ops / Monitoring

- Log recommendations and model/rule version.
- Track feedback such as click, dismiss, or ignore.
- Monitor action distribution and recommendation fatigue.
- Add simple quality checks for explanation faithfulness and safety rules.

11. MVP Scope

Phase 1: Simulated Event Environment

Must-have:

1. Define user personas and behavior scenarios.
2. Define event schema.
3. Define action catalog and recommendation targets.
4. Build synthetic event generation logic.

Phase 2: Next Best Action Engine

Must-have:

1. User feature engineering pipeline.
2. Persona detection.
3. Candidate action generation.
4. Next Best Action scoring/ranking.
5. Eligibility, fatigue, and safety filtering.
6. Explanation for recommendations.
7. Feedback logging.

Phase 3: Lightweight Dashboard & Evaluation

Must-have:

1. User profile view.
2. Recommendation result view.
3. Feedback simulation.
4. Evaluation view.
5. Final report.

Should-have:

1. Clustering-based persona discovery.
2. Simple ML classifier/ranker.
3. Feedback simulation.
4. Fatigue control.
5. Safety rule for finance-related content.

Out of scope:

1. Real user data.
2. Real production super-app.
3. Full mock app UI.

4. Real-time streaming.
5. Deep learning recommendation model.
6. Investment advice or financial product recommendation.
7. Production-level A/B testing.

12. Evaluation Plan

Since real user data and real click labels are not available, evaluation will be based on synthetic scenarios and ranking metrics.

Evaluation methods:

- Scenario accuracy: does the system recommend the expected action for designed user scenarios?
- Precision@K: are the top K recommendations relevant?
- NDCG@K: are better actions ranked higher?
- Persona consistency: do similar users receive similar personas?
- Explanation faithfulness: do explanations match actual user features?
- Fatigue rule check: avoid recommending the same action too often.
- Safety rule check: avoid financial advice or unsafe recommendation.

Example scenario test:

Given:

- voucher_redeemed_30d = 7
- payment_count_30d = 10
- campaign_click_rate = high

Expected:

- cashback campaign should rank in top 1 or top 2.

13. 5-week Plan

Week 1

- Define user personas and behavior scenarios.
- Define event schema and action catalog.
- Build synthetic data generation logic.

Week 2

- Generate synthetic user events.

- Build feature engineering pipeline.
- Implement rule-based baseline recommendation engine.

Week 3

- Add clustering or simple ML ranking/classification model.
- Compare ML results with rule-based baseline.

Week 4

- Build lightweight dashboard.
- Add explanation generation and feedback logging.

Week 5

- Evaluate results.
- Write final report.
- Prepare demo showing user profiles, recommendations, explanations, and evaluation metrics.

14. Risks and Limitations

- Synthetic data may not represent real user behavior.
- The project cannot prove real conversion or engagement improvement.

- The recommendation model should not be treated as production-ready.
- Finance-related actions must be limited to education content, not investment advice.
- The dashboard is only a demo/inspection tool, not a real consumer app.
- The main learning value is the design of a transparent and evaluable Next Best Action pipeline.

15. Questions for Mentor

1. Is it acceptable to use a simulated event/action environment instead of real app data or a full mock app UI?
2. Is this direction suitable for a 5-week AI/DS/ML internship project?
3. Should the scope focus more on recommendation/ranking, user persona, or evaluation?
4. Is synthetic data acceptable if the assumptions and limitations are clearly documented?
5. Should I use a rule-based baseline only, or add simple ML such as clustering/classification?
6. Which output would be most valuable for final review: dashboard, API demo, evaluation report, or all three?

- Dataset
- Methodology - Uber, zalo, ... 2tower(LLM..) kaggle, eda,...
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